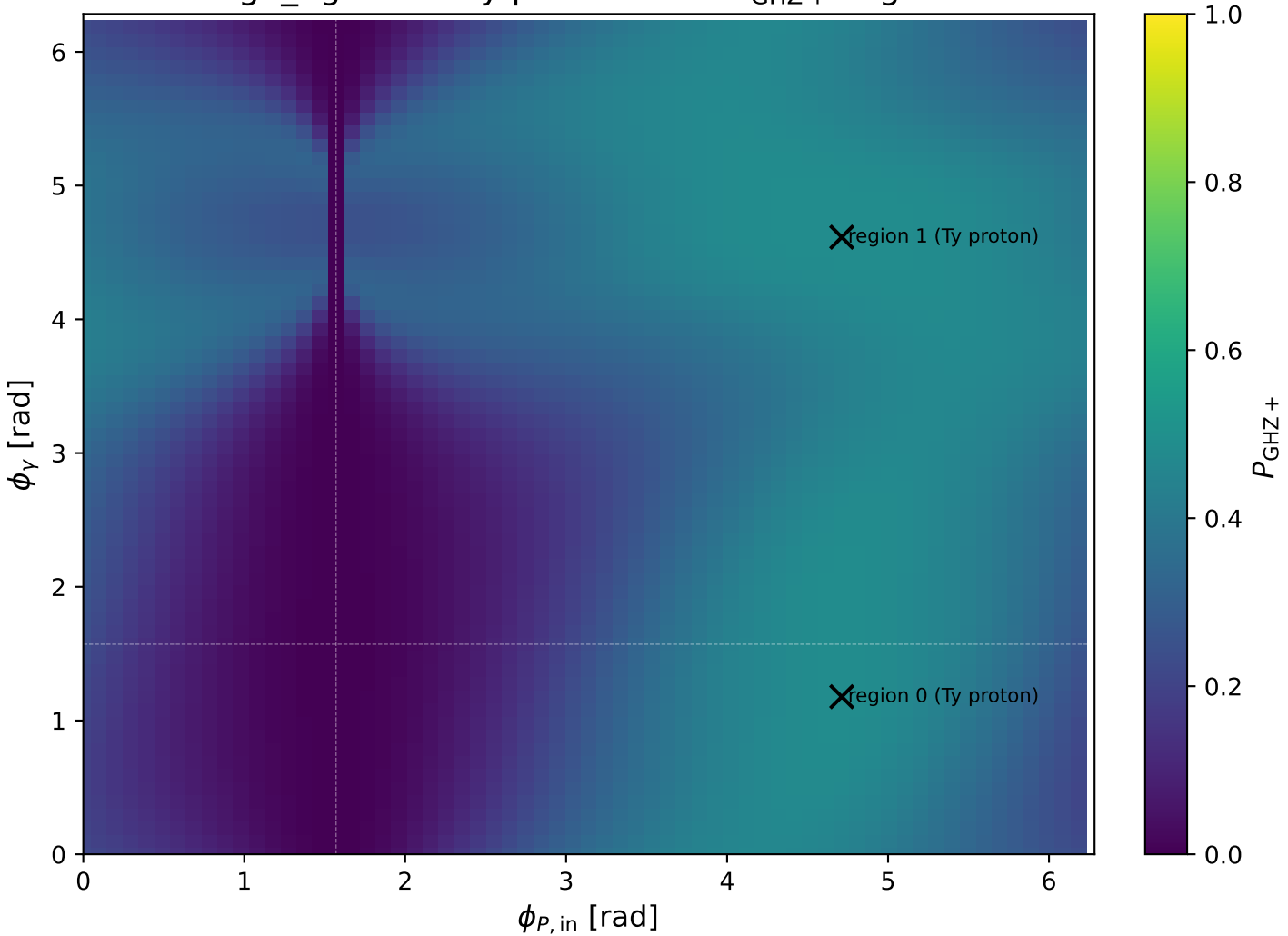


high_Egamma Ty proton: max $P_{\text{GHZ}+}$ regions



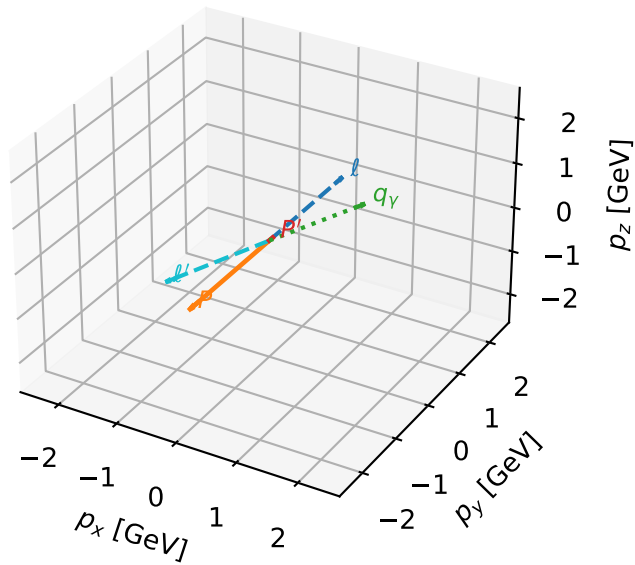
Ty proton characteristic max $P_{\text{GHZ+}}$ configuration

ghz_purity_high_egamma_ty_proton_region_0 $P_{\{\text{rm GHZ+}\}}=0.482713$
region: high_Egamma_Ty_proton_region_0
outgoing-state purity: 0.5
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $\frac{1}{2}\sum_{h_{l_0}}\rho(h_{l_0}, T_{y,p})$

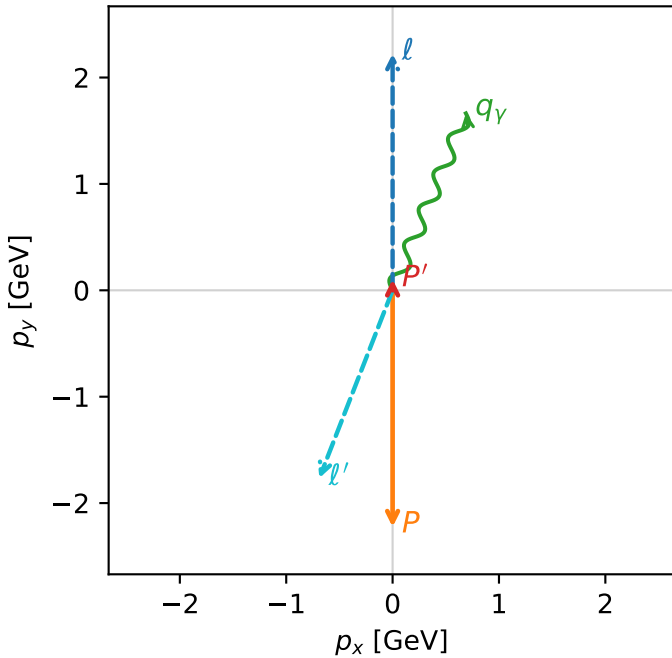
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=0.102337$
 $\theta_{\text{in}}=1.57079$, $\phi_{\ell}=1.5708$, $\phi_P=4.71239$
 $E_{\gamma}=1.8$, $\phi_{\gamma}=1.1781$
 $Q^2=16.2839$, $x_B=0.841965$, $t=-3.25132$, $W^2=3.93631$, $y=0.937217$
 $\theta(\ell', \gamma)=178.816$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= 0.0000$ $p_y= 2.2244$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= -0.0000$ $p_y= -2.2244$ $p_z= 0.0000$
 ℓ' $E= 1.8950$ $p_x= -0.6888$ $p_y= -1.7653$ $p_z= 0.0000$
 P' $E= 0.9436$ $p_x= 0.0000$ $p_y= 0.1023$ $p_z= 0.0000$
 q_{γ} $E= 1.8000$ $p_x= 0.6888$ $p_y= 1.6630$ $p_z= 0.0000$

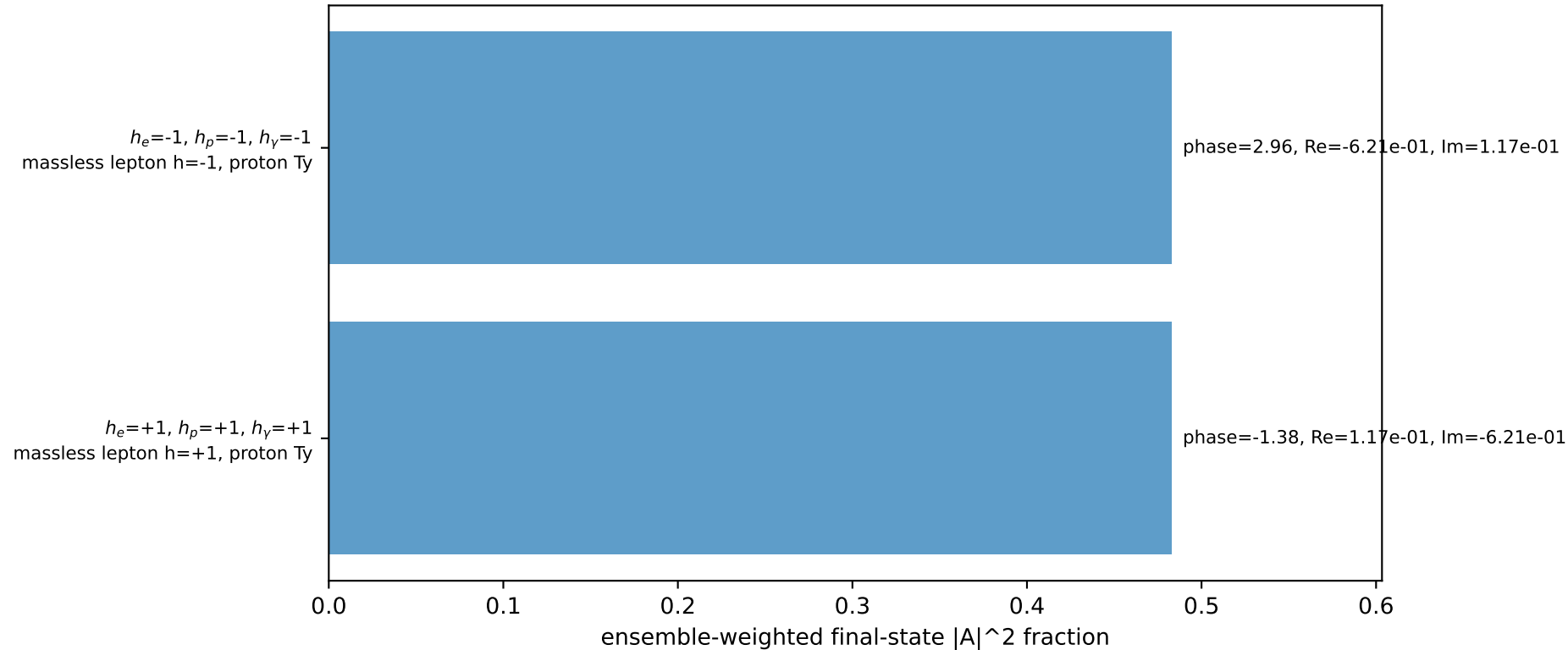
3D momenta



Transverse plane

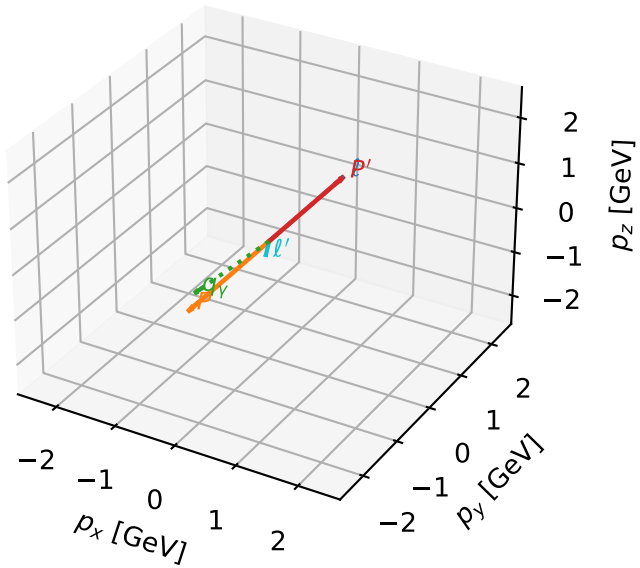


Leading initial-ensemble/final-state components (N=2, retained=96.5%)



Ty proton characteristic max $P_{\text{GHZ+}}$ configuration

3D momenta

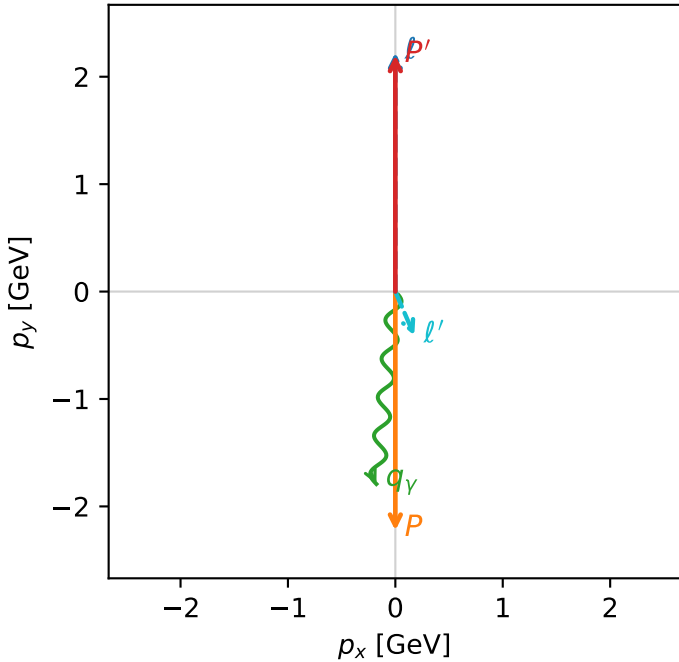


ghz_purity_high_egamma_ty_proton_region_1 $P_{\{\text{rm GHZ+}\}}=0.482463$
 region: high_Egamma_Ty_proton_region_1
 outgoing-state purity: 0.5
 kinematic point: high_s_high_theta_in_high_Egamma
 incoming state: $\frac{1}{2}\sum_{h_{\ell_0}}\rho(h_{\ell_0}, T_{y,p})$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=2.20096$
 $\theta_{\text{in}}=1.57079$, $\phi_{\ell}=1.5708$, $\phi_P=4.71239$
 $E_{\gamma}=1.8$, $\phi_{\gamma}=4.61421$
 $Q^2=3.80661$, $x_B=0.187472$, $t=-19.5835$, $W^2=17.3782$, $y=0.983958$
 $\theta(\ell', \gamma)=28.9268$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= 0.0000$ $p_y= 2.2244$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= -0.0000$ $p_y= -2.2244$ $p_z= 0.0000$
 ℓ' $E= 0.4460$ $p_x= 0.1764$ $p_y= -0.4096$ $p_z= 0.0000$
 P' $E= 2.3925$ $p_x= 0.0000$ $p_y= 2.2010$ $p_z= 0.0000$
 q_{γ} $E= 1.8000$ $p_x= -0.1764$ $p_y= -1.7913$ $p_z= 0.0000$

Transverse plane



$h_e=-1$, $h_p=-1$, $h_{\gamma}=-1$
 massless lepton $h=-1$, proton Ty

$h_e=+1$, $h_p=+1$, $h_{\gamma}=+1$
 massless lepton $h=+1$, proton Ty

Leading initial-ensemble/final-state components (N=2, retained=96.5%)

